

Digital System Design

LAB II

Testbench and simulation

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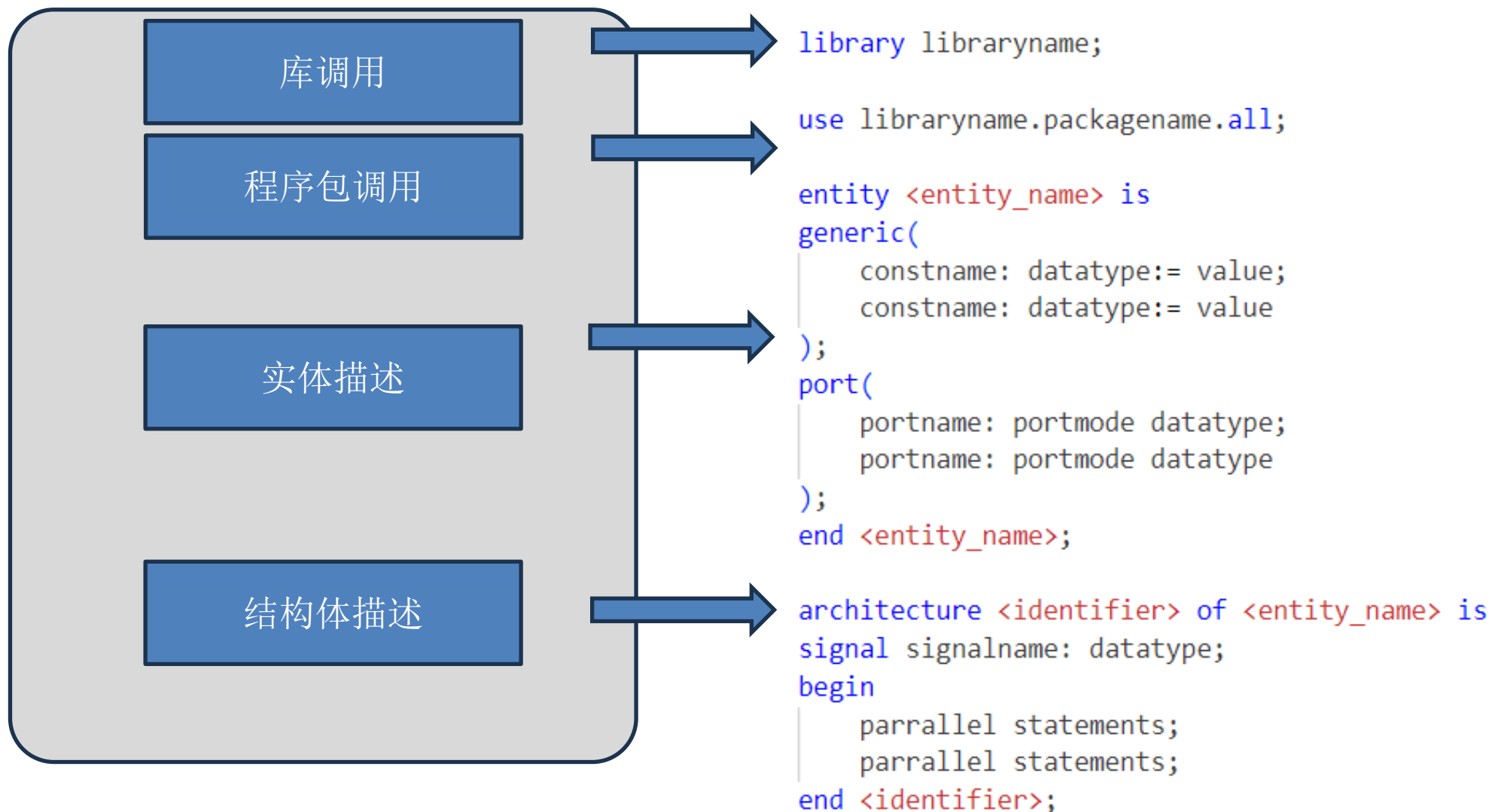
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Review - VHDL

VHDL基本结构

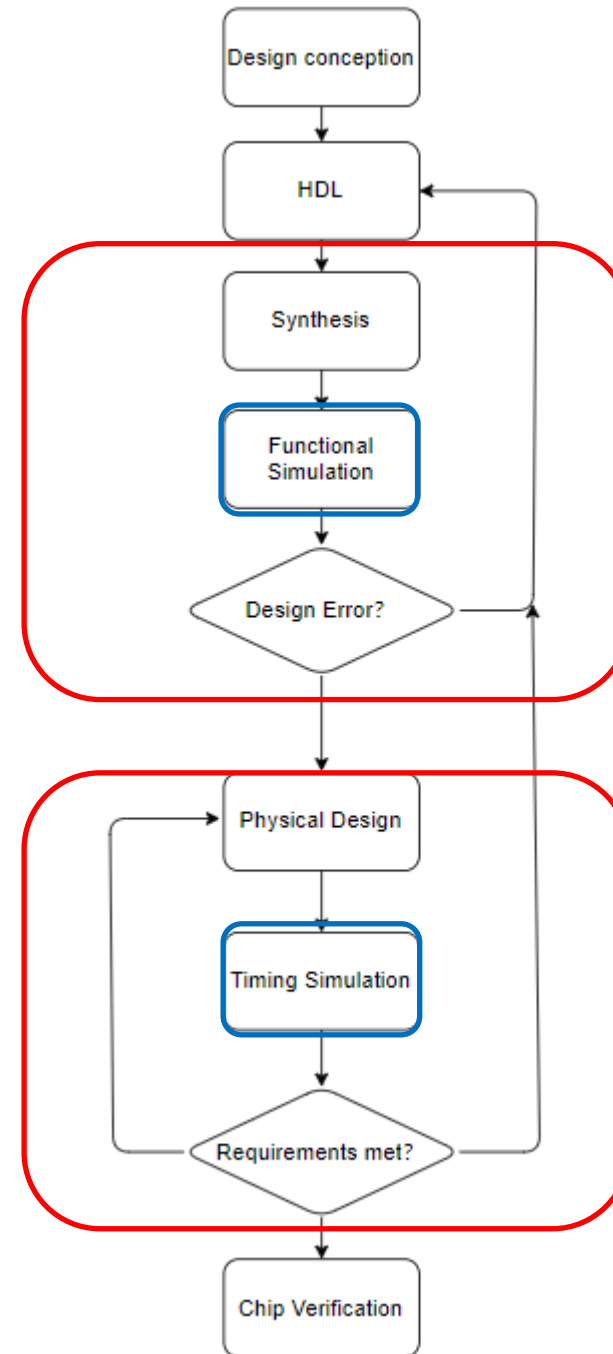


Review - EDA Flowchart

RTL Analysis
Synthesis

Implementation

Program and Debug



VHDL Basic

- 库和程序包的使用
- LIBRARY IEEE
- USE IEEE_Std_Logic_1164.all
 - 定义了标准逻辑类型std_logic / std_logic_vector以及相关的类型转换函数

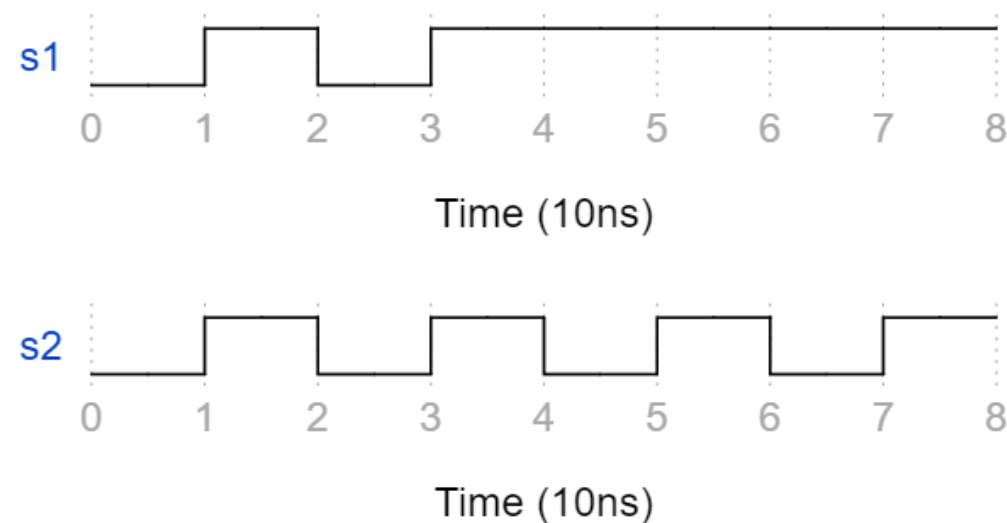
VHDL Basic

- 逻辑运算

操作符	功能
NOT	非
AND	与
OR	或
XOR	异或
NAND	与非
NOR	或非
XNOR	同或

VHDL Basic

- 信号赋值语句
- 非周期信号
- `s1 <= '0', '1' after 10ns, '0' after 20ns, '1' after 30ns`
- 周期信号
- `signal s2: std_logic := '0'`
- `s2 <= not s2, after 10ns`
- 延迟语句只对仿真器有效
- 绝大部分综合器不会考虑延迟



VHDL Basic

- 例化语句
- 结构体说明区域中进行声明
- 结构体内进行例化

```
architecture <identifier> of <entity_name_tb> is
```

```
component <entity_name> is is 在这里可写可不写
```

```
    Port (  
        portname1: portmode datatype;  
        portname2: portmode datatype  
    );  
end component;
```

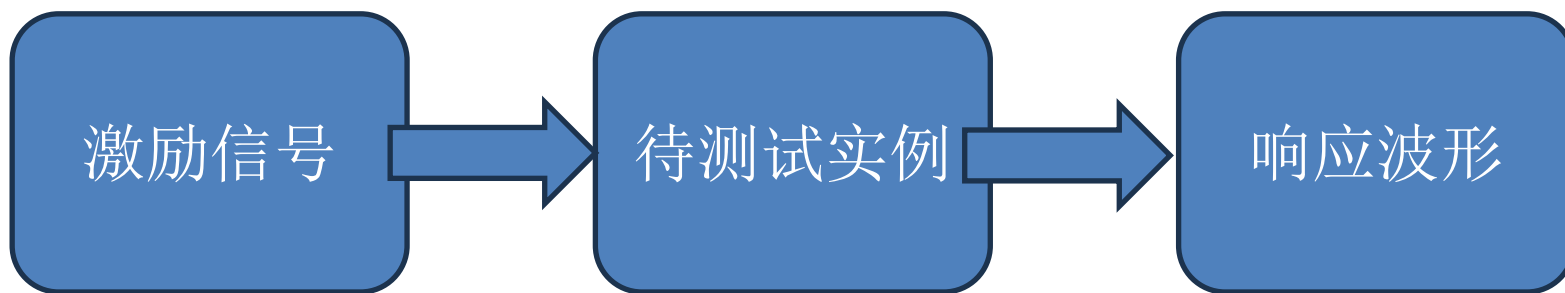
```
begin  
    <instance_name> : <entity_name>  
        port map (  
            portname1 => signalname1,  
            portname2 => signalname2  
        );  
end
```

```
<identifier>;
```

<端口名> => <信号名>

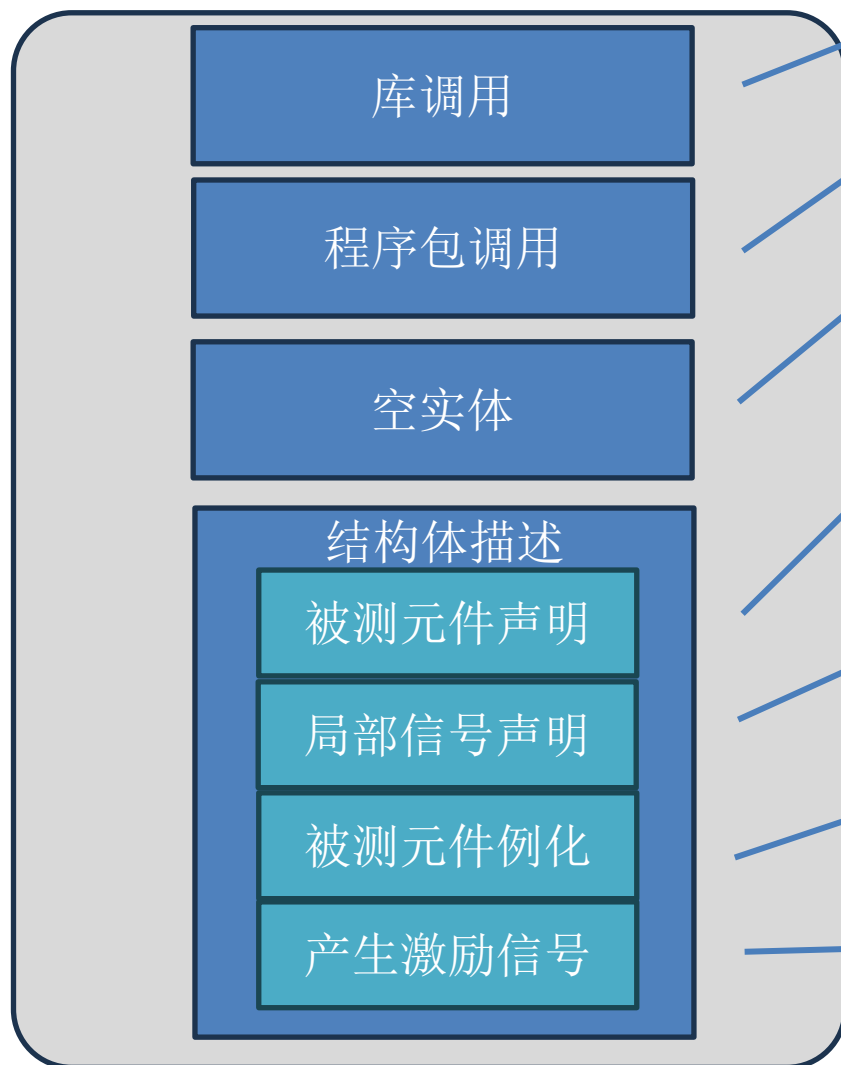
Testbench

- 编写Testbench对使用硬件描述语言（HDL）设计的电路系统进行仿真验证，测试设计系统功能或部分性能是否与预期的目标相符。

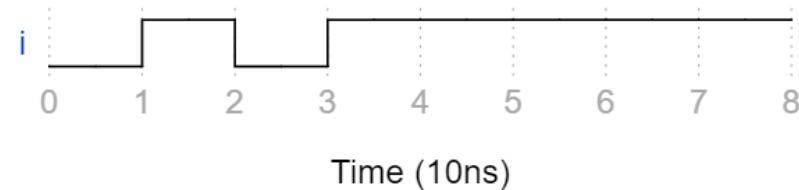


Testbench

Testbench基本结构

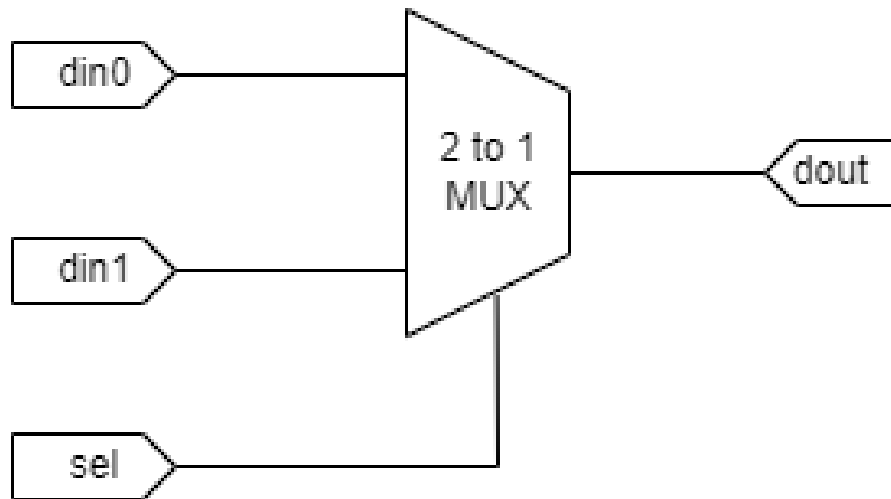


```
library ieee;  
use ieee.std_logic_1164.all;  
  
entity led_tb is  
end led_tb;  
  
architecture tb of led_tb is  
  component led  
    port (  
      input : in std_logic;  
      output : out std_logic  
    );  
  end component;  
  
  signal i,o,clk: std_logic;  
  constant time_unit: time := 10 ns;  
  
begin  
  DUT : led port  
    map (  
      input => i,  
      output => o  
    );  
  
  i <= '0', '1' after time_unit, '0' after time_unit*2, '1' after time_unit*3;  
  
end tb;
```



2-to-1 Multiplexer

设计一个二选一多路选择器

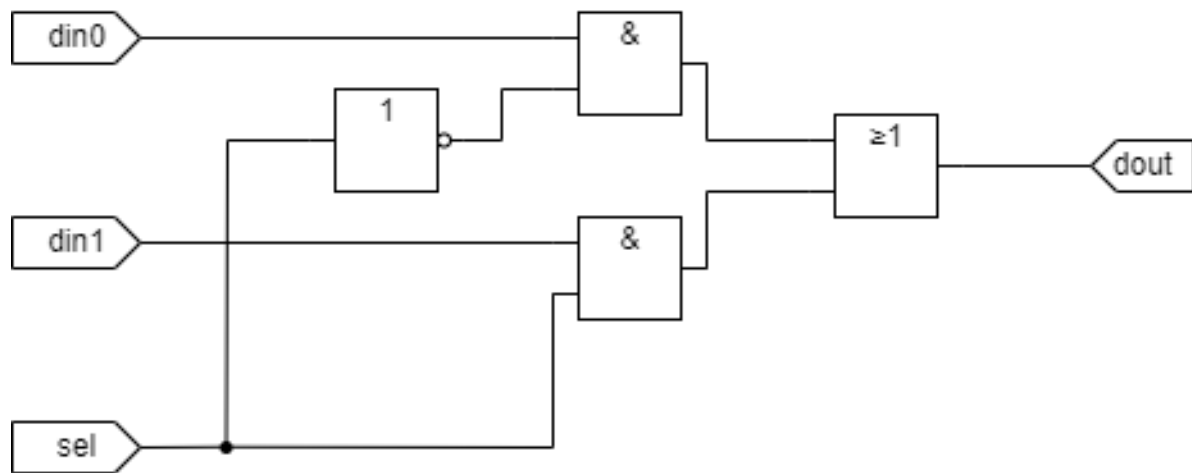


din0	din1	sel	dout
0	0	0	0
1	0	0	1
0	1	0	0
1	1	0	1
0	0	1	0
1	0	1	0
0	1	1	1
1	1	1	1

2-to-1 Multiplexer

二选一多路选择器

$$dout = din0 \cdot \overline{sel} + din1 \cdot sel$$



```
library ieee;
use ieee.std_logic_1164.all;

entity mux2_1 is
    port(
        din0: in std_logic;
        din1: in std_logic;
        sel:  in std_logic;
        dout: out std_logic
    );
end mux2_1;

architecture dataflow of mux2_1 is
begin
    dout <= (din0 and not sel) or (din1 and sel);
end dataflow;
```

2-to-1 Multiplexer

测试文件

```
library ieee;
use ieee.std_logic_1164.all;

--空实体
entity mux2_1_tb is

end mux2_1_tb;

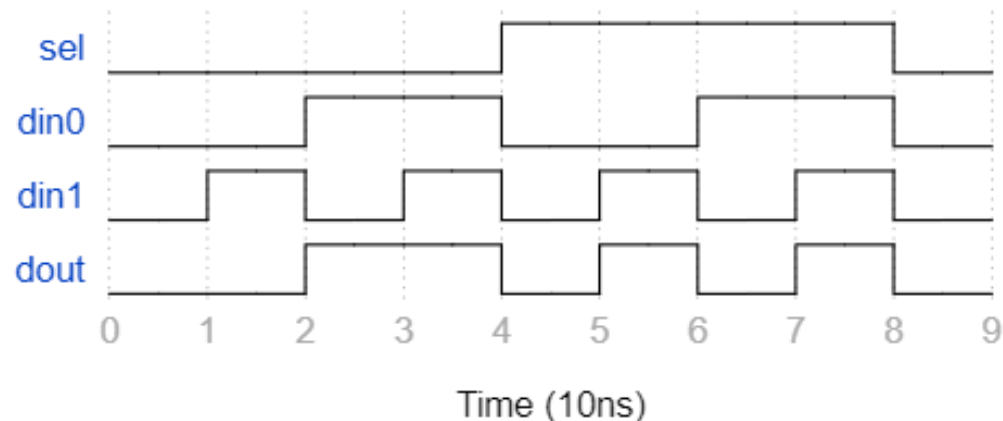
--结构体
architecture tb of mux2_1_tb is
--元件声明
component mux2_1 is
    port(
        din0: in std_logic;
        din1: in std_logic;
        sel: in std_logic;
        dout: out std_logic
    );
end component;
```

```
--信号声明
signal d0, d1, s, do: std_logic;

--常数声明
constant period: time := 10 ns;

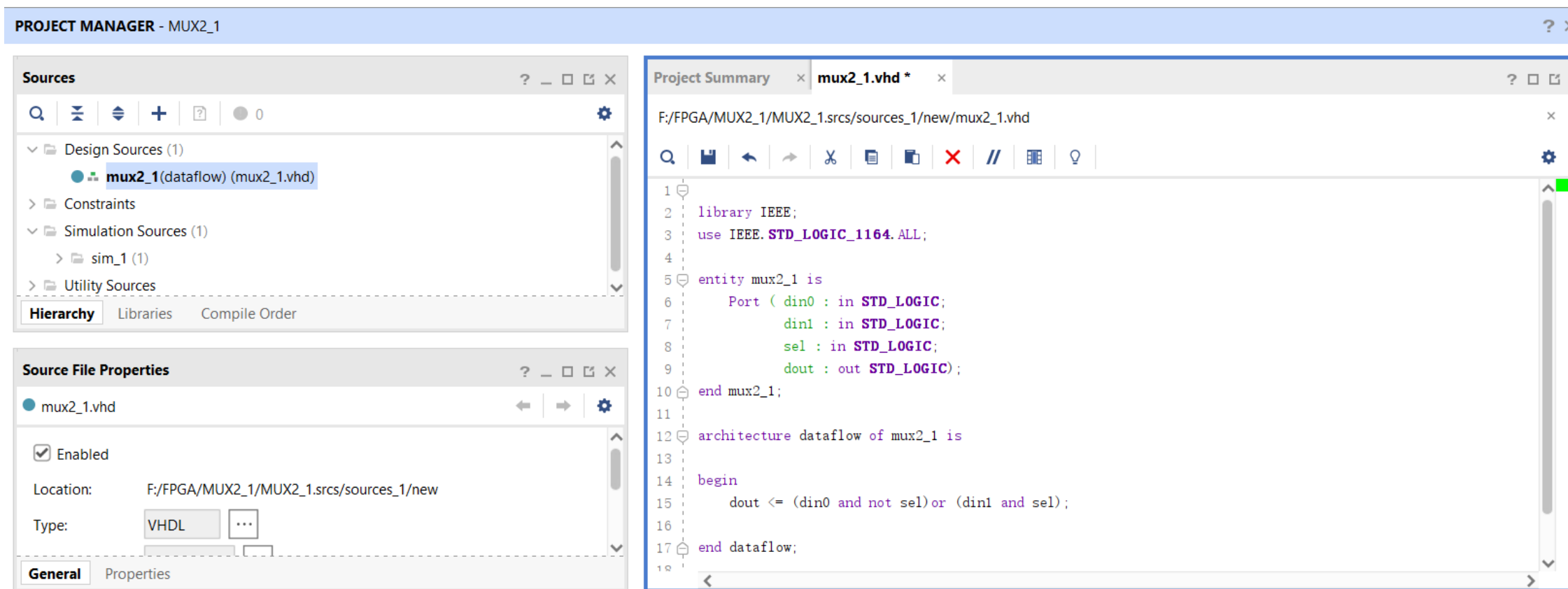
begin
    --实例化
    uut: mux2_1 port map(din0 => d0, din1 => d1, sel => s, dout => do);
    --测试用例
    s <= '0', '1' after period*4, '0' after period*8;
    d0 <= '0', '1' after period*2,
        '0' after period*4, '1' after period*6, '0' after period*8;
    d1 <= '0', '1' after period,
        '0' after period*2, '1' after period*3,
        '0' after period*4, '1' after period*5,
        '0' after period*6, '1' after period*7, '0' after period*8;

end tb;
```

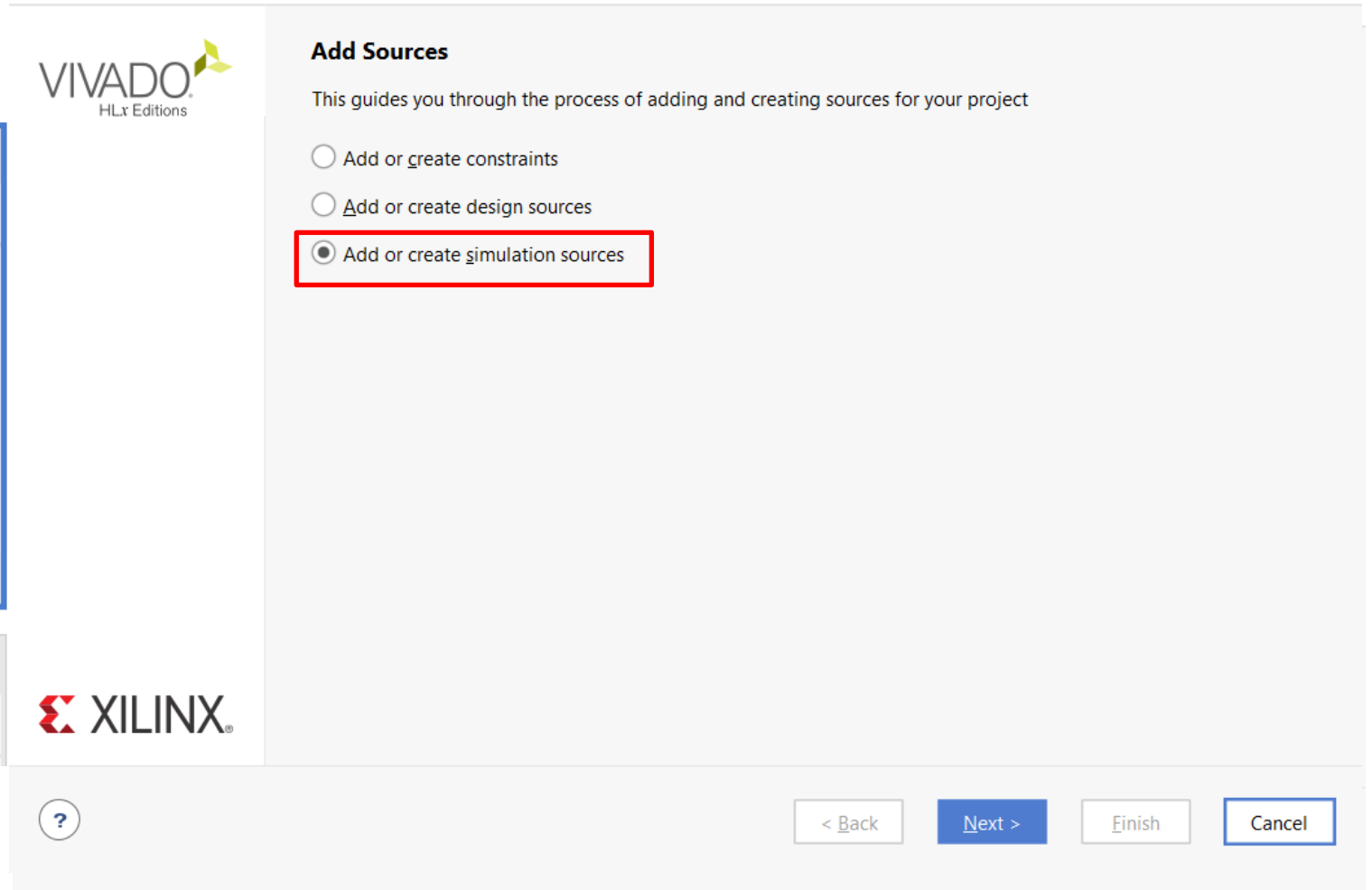
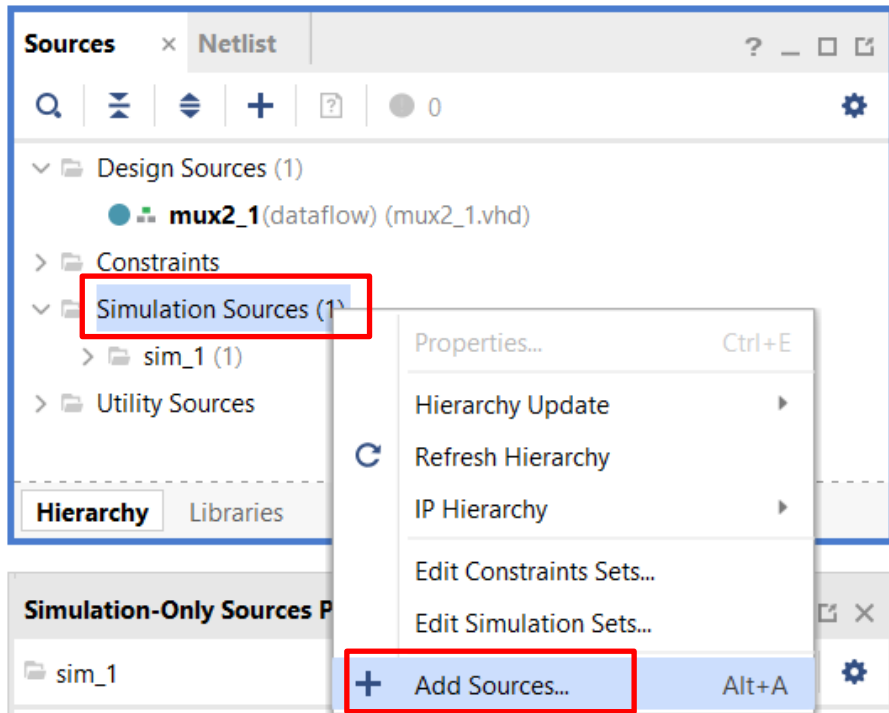


Create Design Source

创建工程，添加设计文件



Create Design Source



Create Simulation Source

Add or Create Simulation Sources

Specify simulation specific HDL files, or directories containing HDL files, to add to your project. Create a new source file on disk and add it to your project.

Specify simulation set: sim_1

+ | - | ↑ | ↓

Use Add Files, Add Directories or Create File buttons below

Add Files Add Directories Create File

☐ Scan and add RTL include files into project

☒ Copy sources into project

☒ Add sources from subdirectories

☒ Include all design sources for simulation

? < Back Next > Finish Cancel

Create Source File

Create a new source file and add it to your project.

File type: VHDL

File name: mux2_1_tb

File location: <Local to Project>

? OK Cancel

Create Simulation Source

Add or Create Simulation Sources

Specify simulation specific HDL files, or directories containing HDL files, to add to your project. Create a new source file on disk and add it to your project.

Specify simulation set: sim_1

	Index	Name	Library	Location
●	1	mux2_1_tb.vhd	xil_defaultlib	<Local to Project>

☐ Scan and add RTL include files into project

☒ Copy sources into project

☒ Add sources from subdirectories

☒ Include all design sources for simulation

Define Module

Define a module and specify I/O Ports to add to your source file.

For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

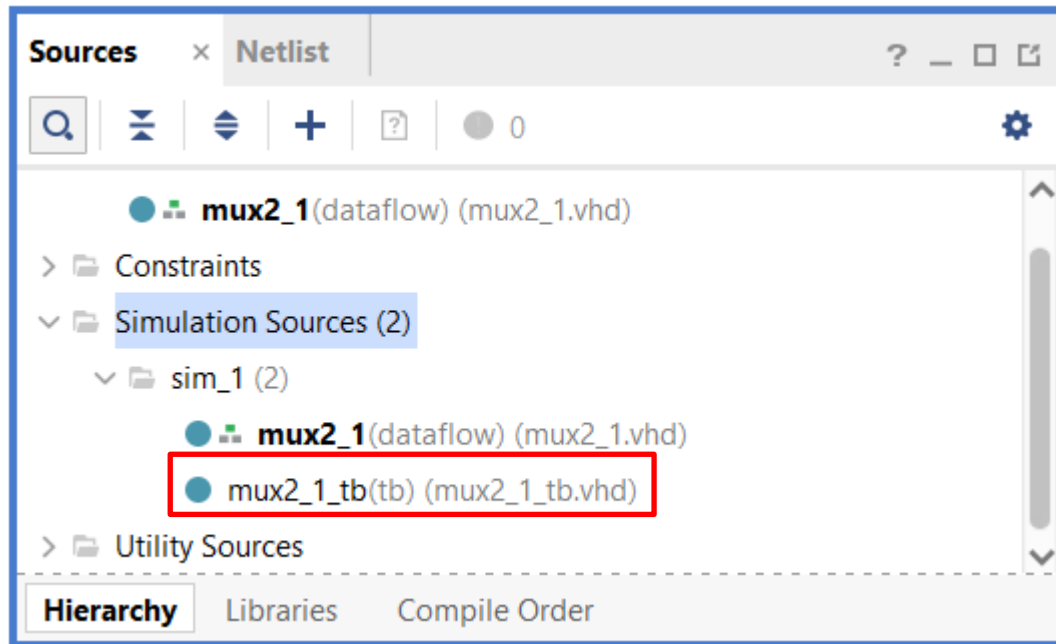
Entity name: mux2_1_tb

Architecture name: tb

I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
	in	☐	0	0

Create Simulation Source



```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity mux2_1_tb is  
-- Port ();  
end mux2_1_tb;
```

```
architecture tb of mux2_1_tb is  
component mux2_1 is  
    Port ( din0 : in STD_LOGIC;  
          din1 : in STD_LOGIC;  
          sel  : in STD_LOGIC;  
          dout : out STD_LOGIC);  
end component;  
signal d0, d1, s, do : STD_LOGIC;  
constant period: time := 10ns;
```

```
begin  
    uut: mux2_1 port  
        map( din0 => d0,  
             din1 => d1,  
             sel  => s,  
             dout => do);  
  
    s <= '0', '1' after period*4, '0' after period*8;  
    d0 <= '0', '1' after period*2,  
         '0' after period*4, '1' after period*6, '0' after period*8;  
    d1 <= '0', '1' after period,  
         '0' after period*2, '1' after period*3,  
         '0' after period*4, '1' after period*5,  
         '0' after period*6, '1' after period*7, '0' after period*8;  
  
end tb;
```

Behavioral Simulation

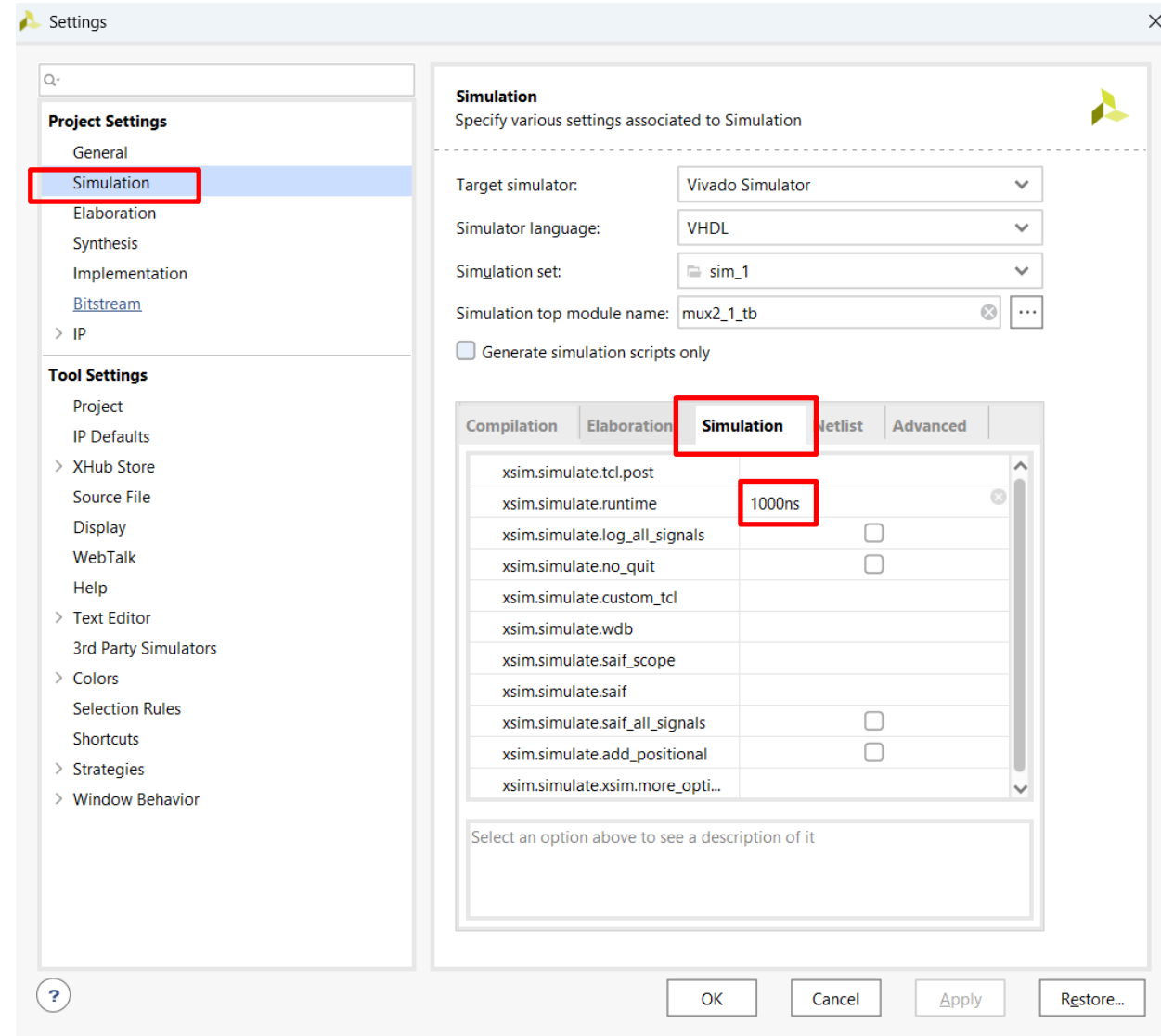
PROJECT MANAGER

Settings

Add Sources

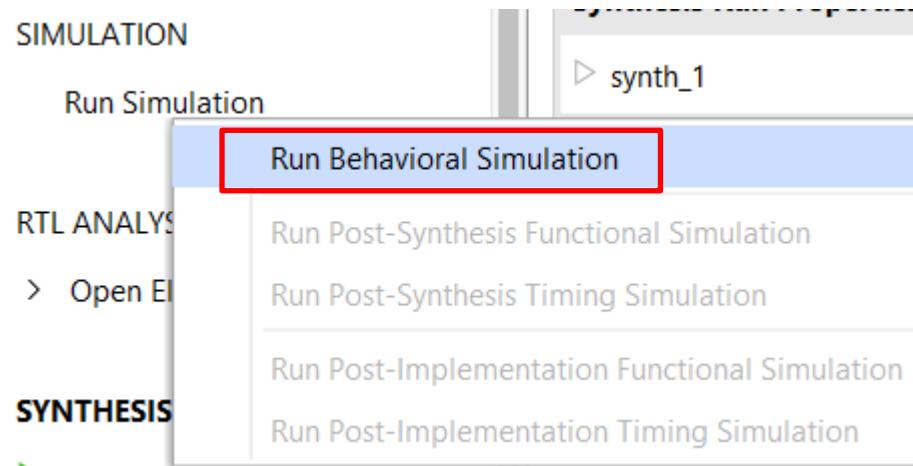
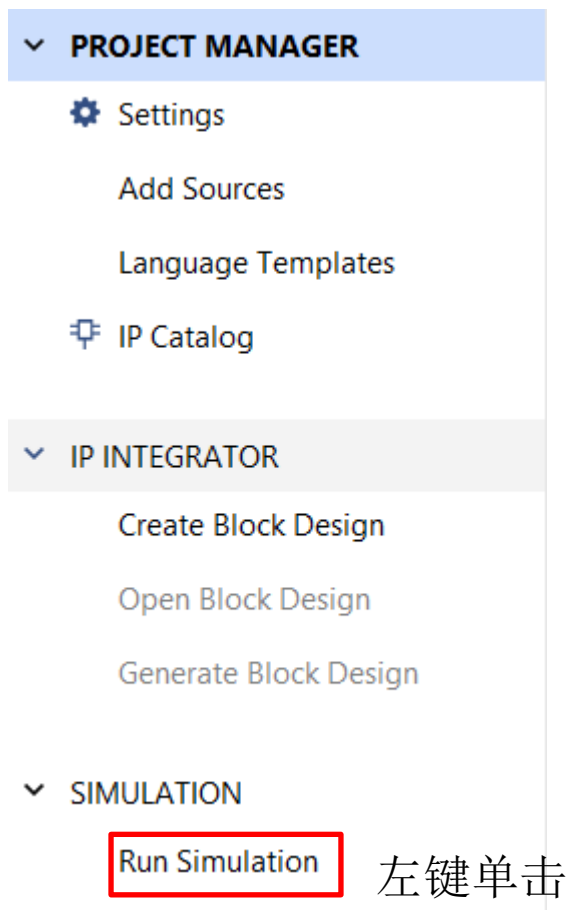
Language Templates

IP Catalog

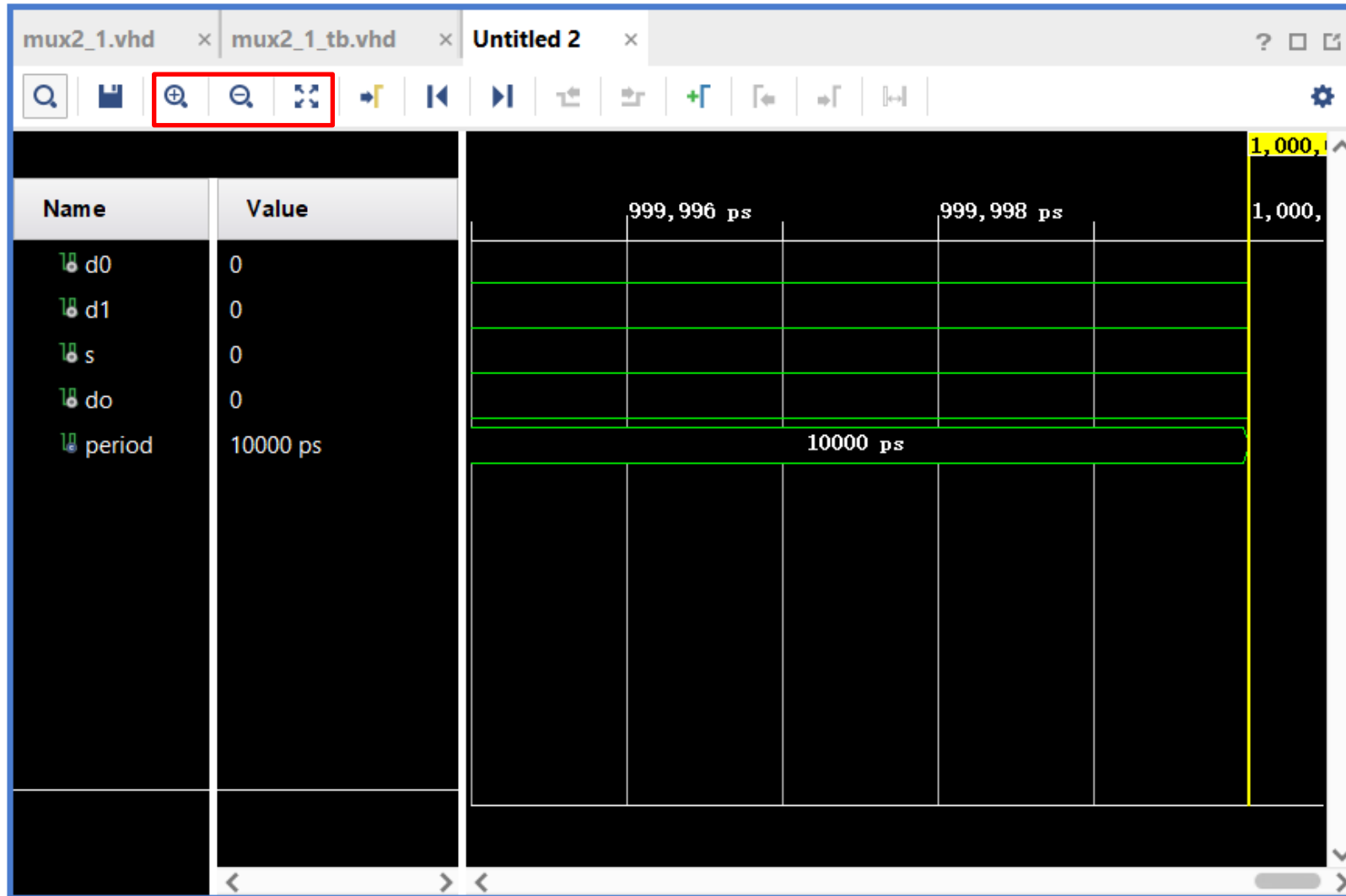


可以将时间改短一些
以提高仿真速度

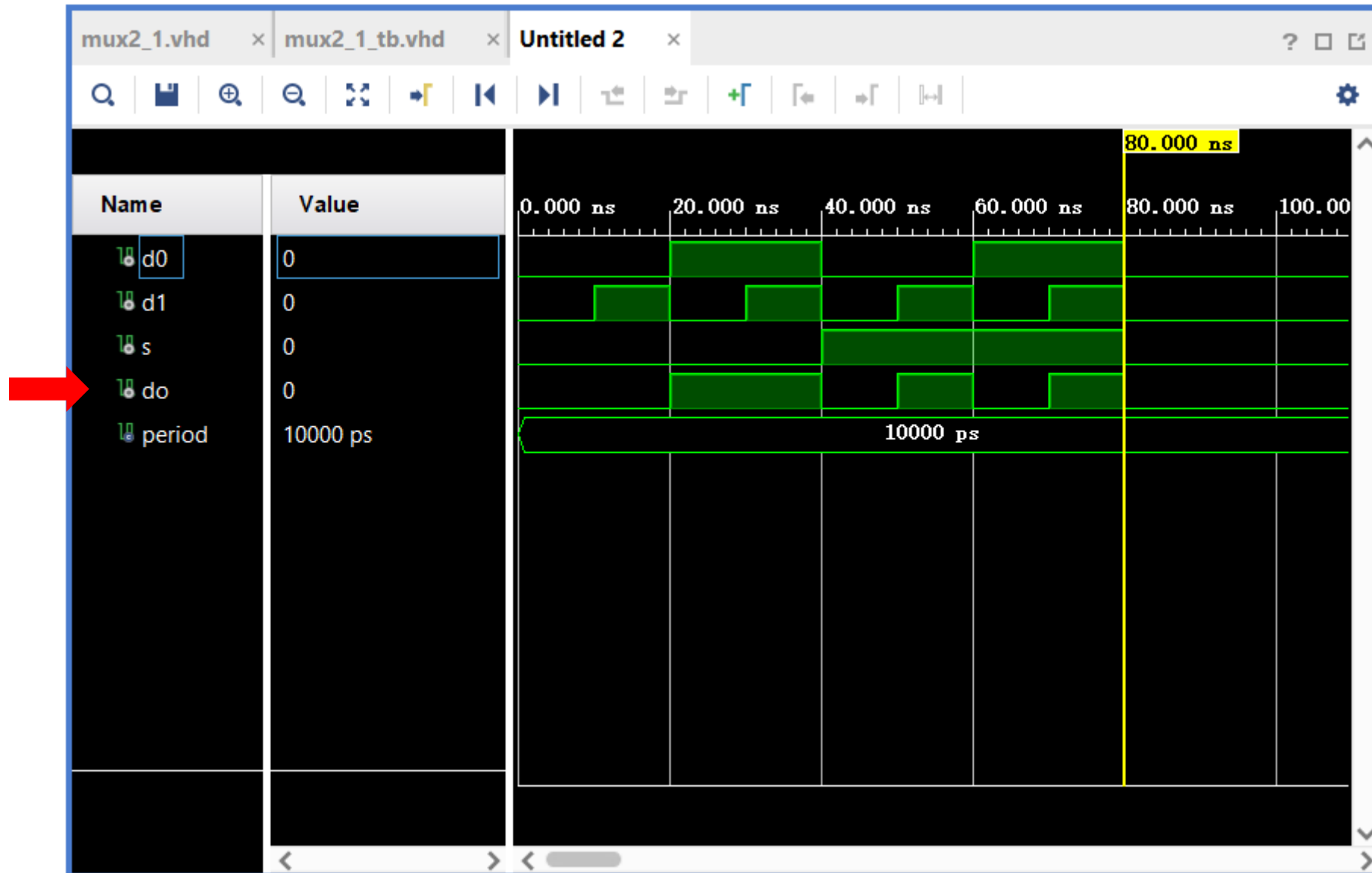
Behavioral Simulation



Behavioral Simulation



Behavioral Simulation



Synthesis

Run Synthesis

SYNTHESIS

▶ Run Synthesis

Open Synthesized Design

Constraints Wizard

Edit Timing Constraints

Set Up Debug

Report Timing Summary

Report Clock Networks

Report Clock Interaction

Report Methodology

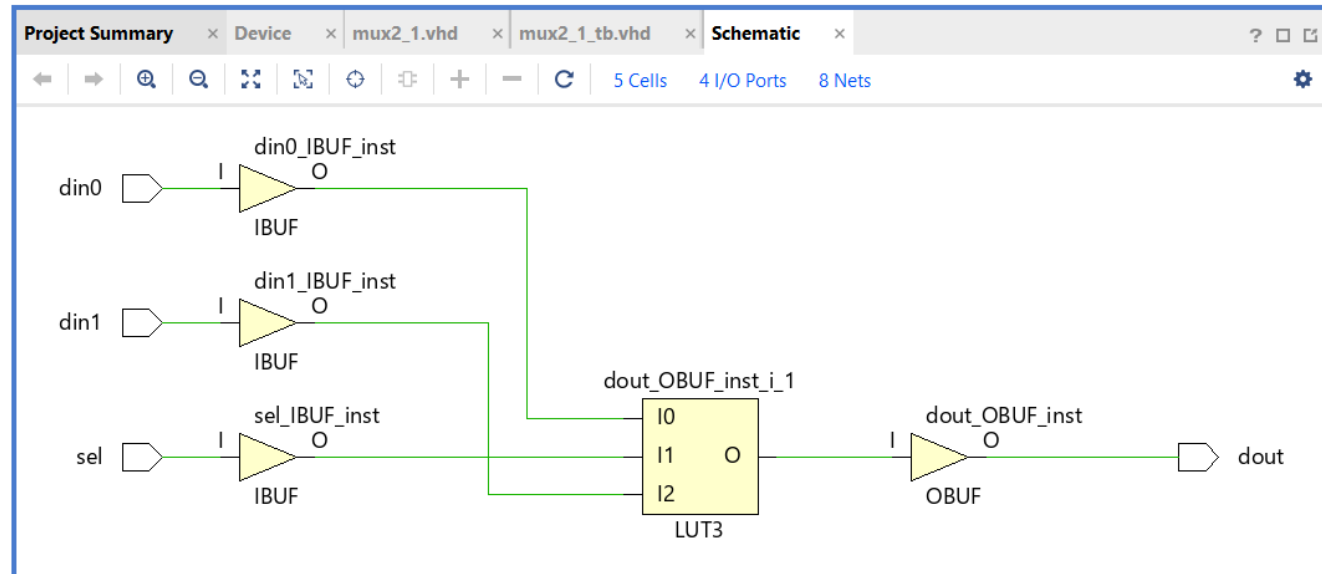
Report DRC

Report Noise

Report Utilization

Report Power

Schematic



Project Summary | Device | mux2_1.vhd | mux2_1_tb.vhd | Schematic

Overview | Dashboard

Strategy: [Vivado Synthesis Defaults](#) | Strategy: [Vivado Implementation Defaults](#)
Report Strategy: [Vivado Synthesis Default Reports](#) | Report Strategy: [Vivado Implementation Default Reports](#)
Incremental synthesis: [None](#) | Incremental implementation: [None](#)

DRC Violations | **Timing**

[Run Implementation](#) to see DRC results | [Run Implementation](#) to see timing results

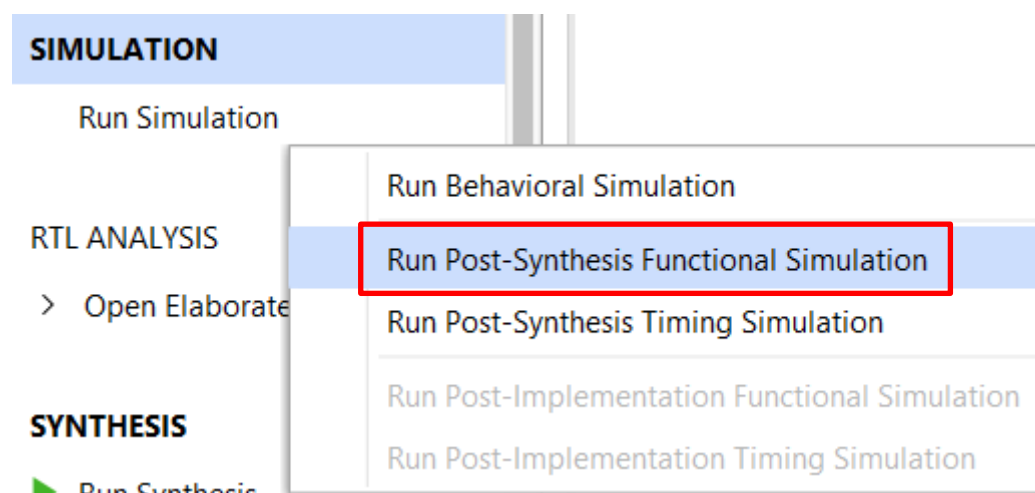
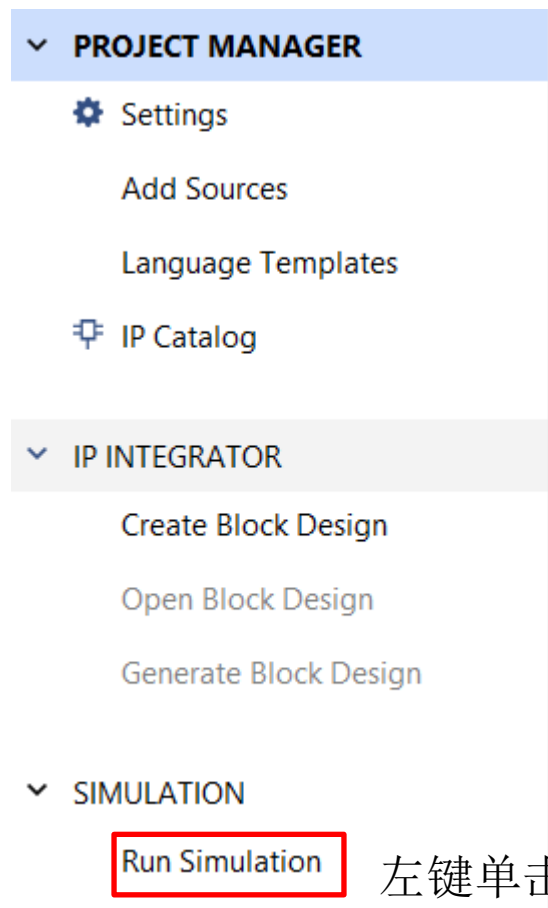
Utilization | **Post-Synthesis** | Post-Implementation | **Power**

Graph | Table

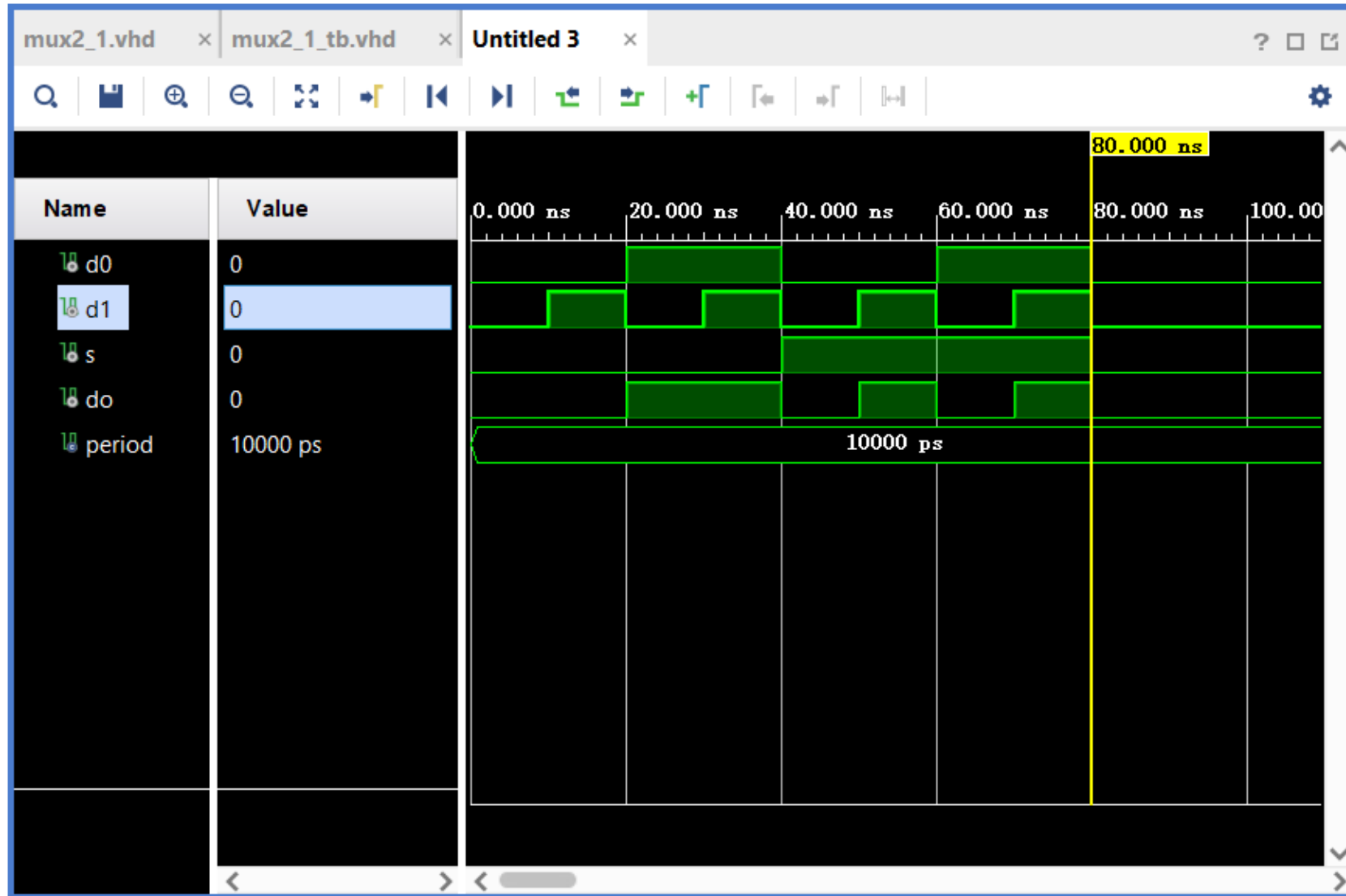
Resource	Estimation	Available	Utilization %
LUT	1	63400	0.01
IO	4	210	1.90

[Run Implementation](#) to see power results

Post-Synthesis Functional Simulation

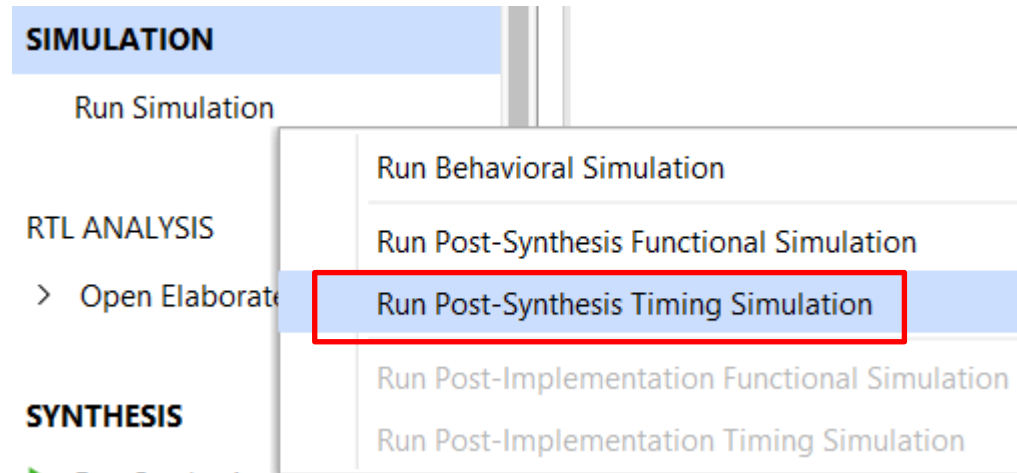
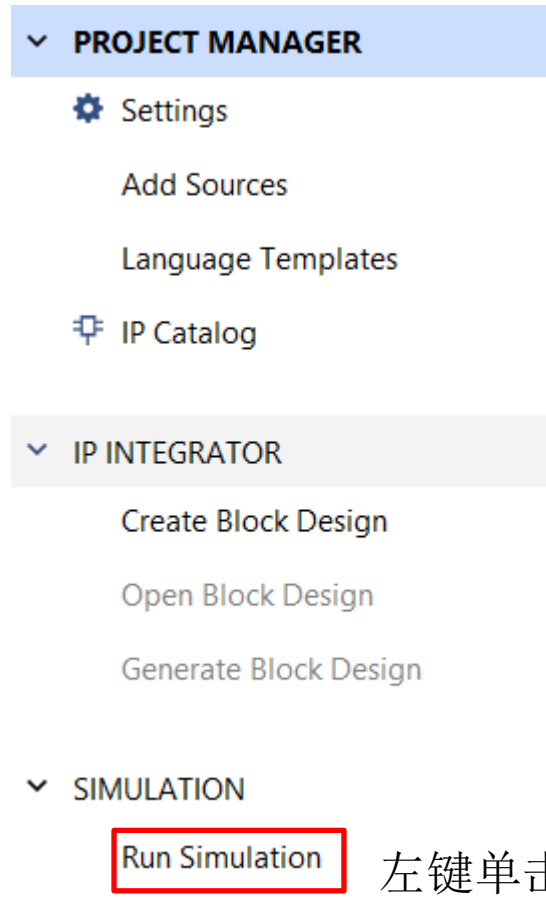


Post-Synthesis Functional Simulation

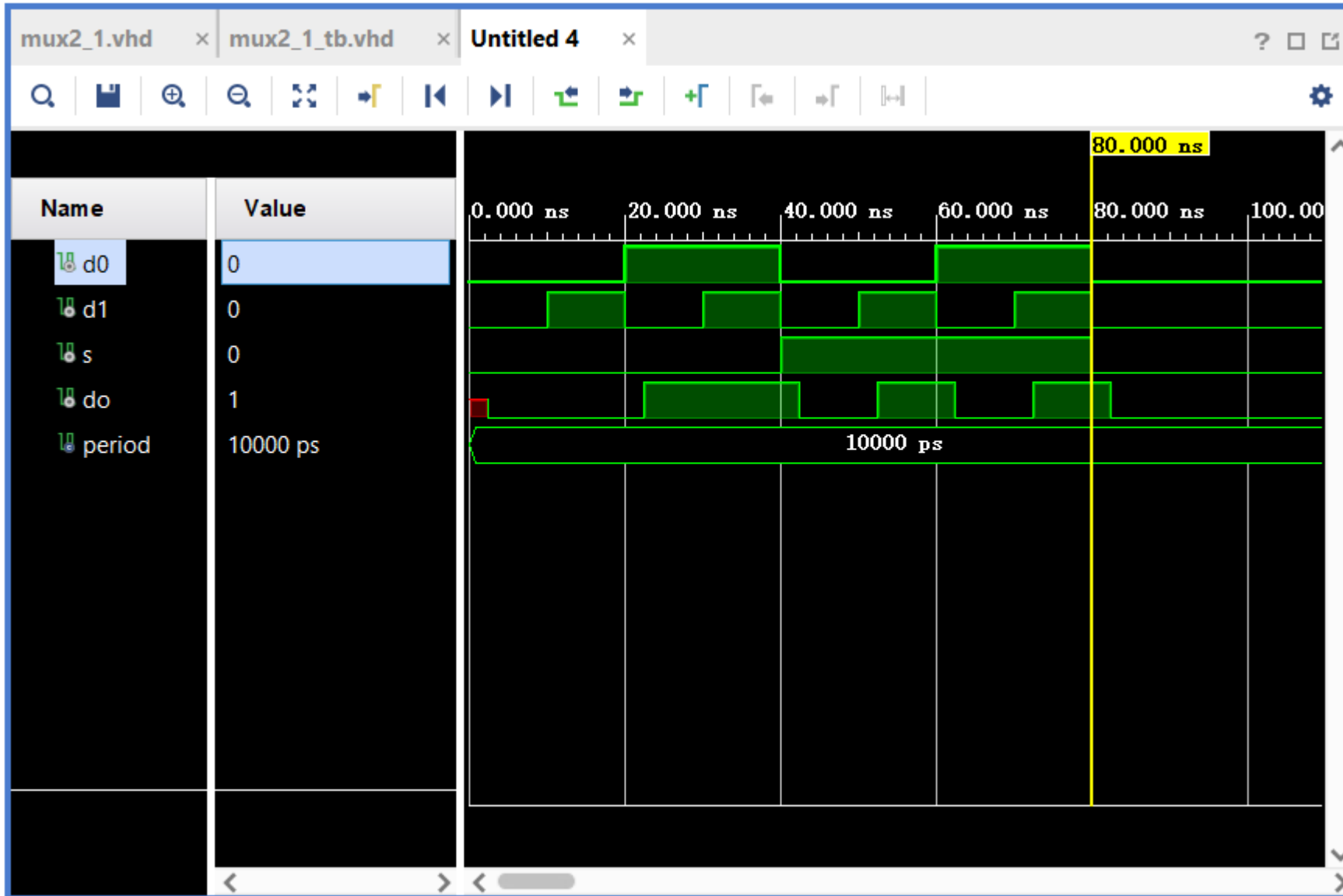


功能仿真与行为仿真一致

Post-Synthesis Timing Simulation

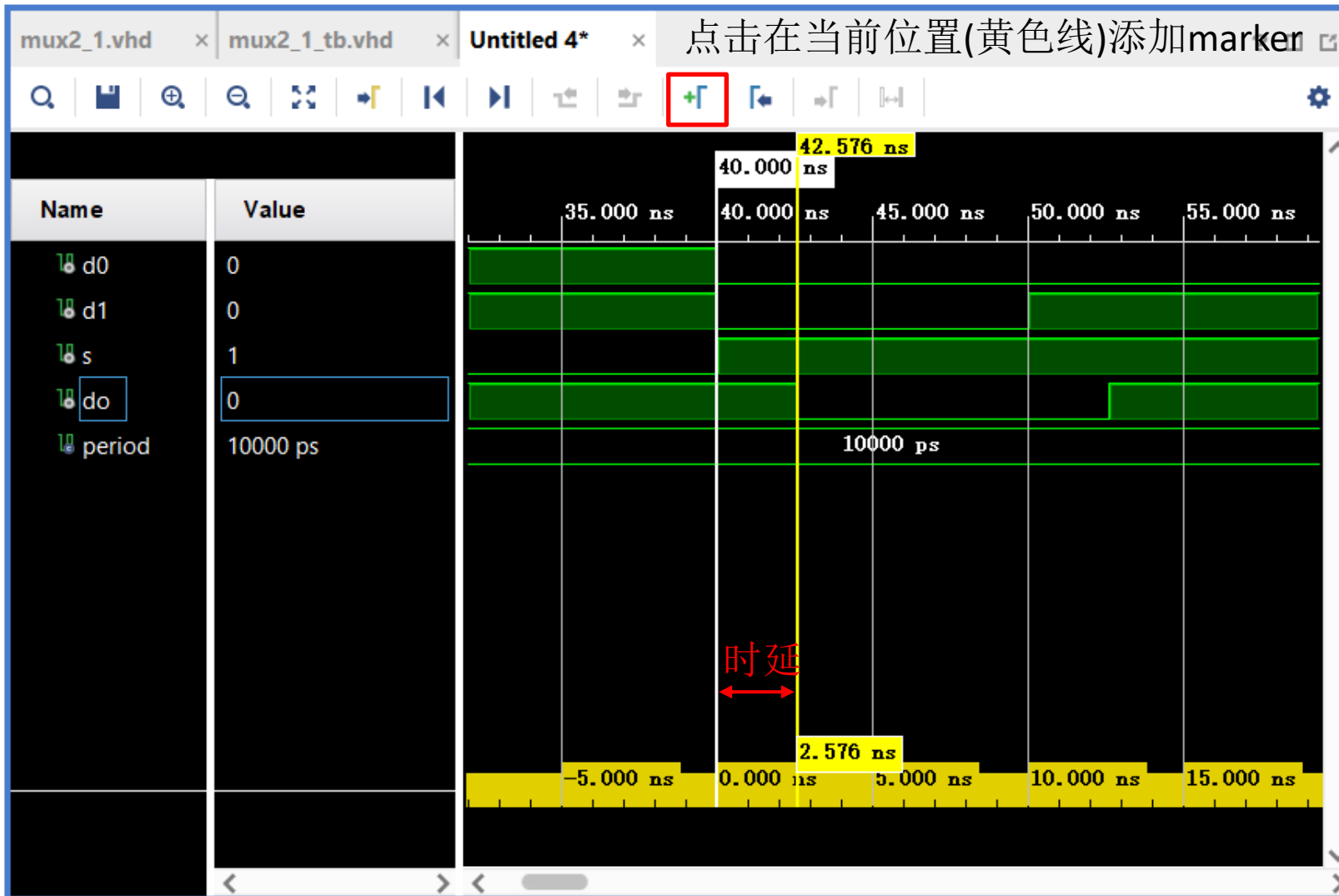


Post-Synthesis Timing Simulation



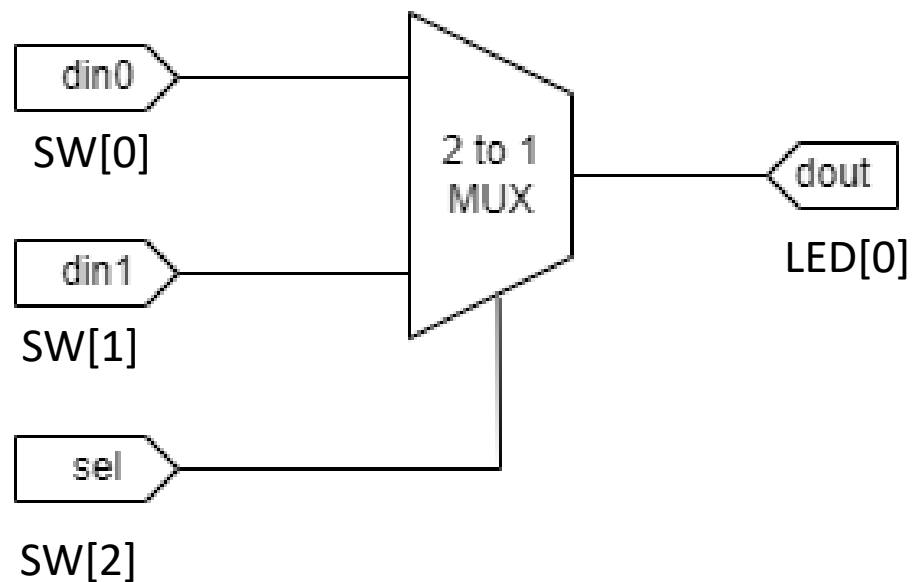
时序仿真的输出会考虑门延迟

Post-Synthesis Timing Simulation



点击marker线更改坐标

Constraint File



13-15行

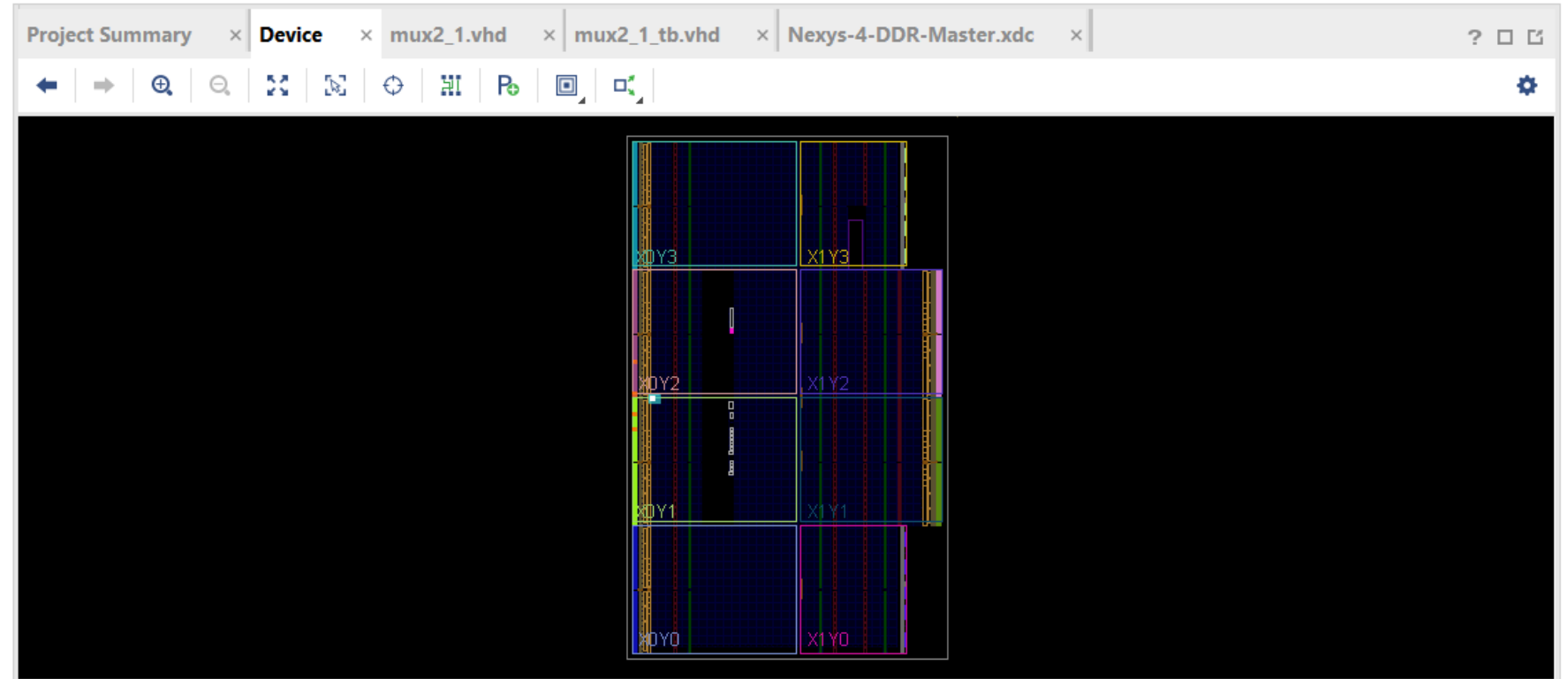
```
set_property -dict { PACKAGE_PIN J15    IOSTANDARD LVCMOS33 } [get_ports { din0 }];  
set_property -dict { PACKAGE_PIN L16    IOSTANDARD LVCMOS33 } [get_ports { din1 }];  
set_property -dict { PACKAGE_PIN M13    IOSTANDARD LVCMOS33 } [get_ports { sel }];
```

33行

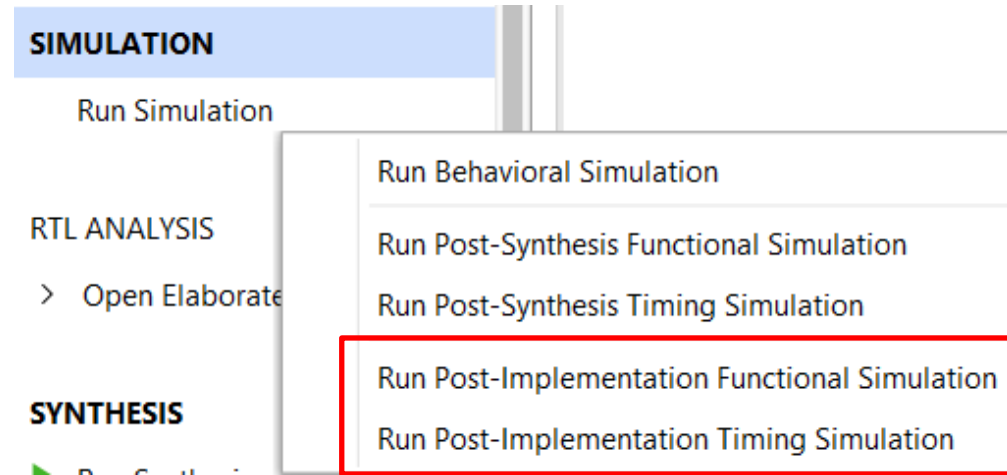
```
set_property -dict { PACKAGE_PIN H17    IOSTANDARD LVCMOS33 } [get_ports { dout }];
```

Implementation

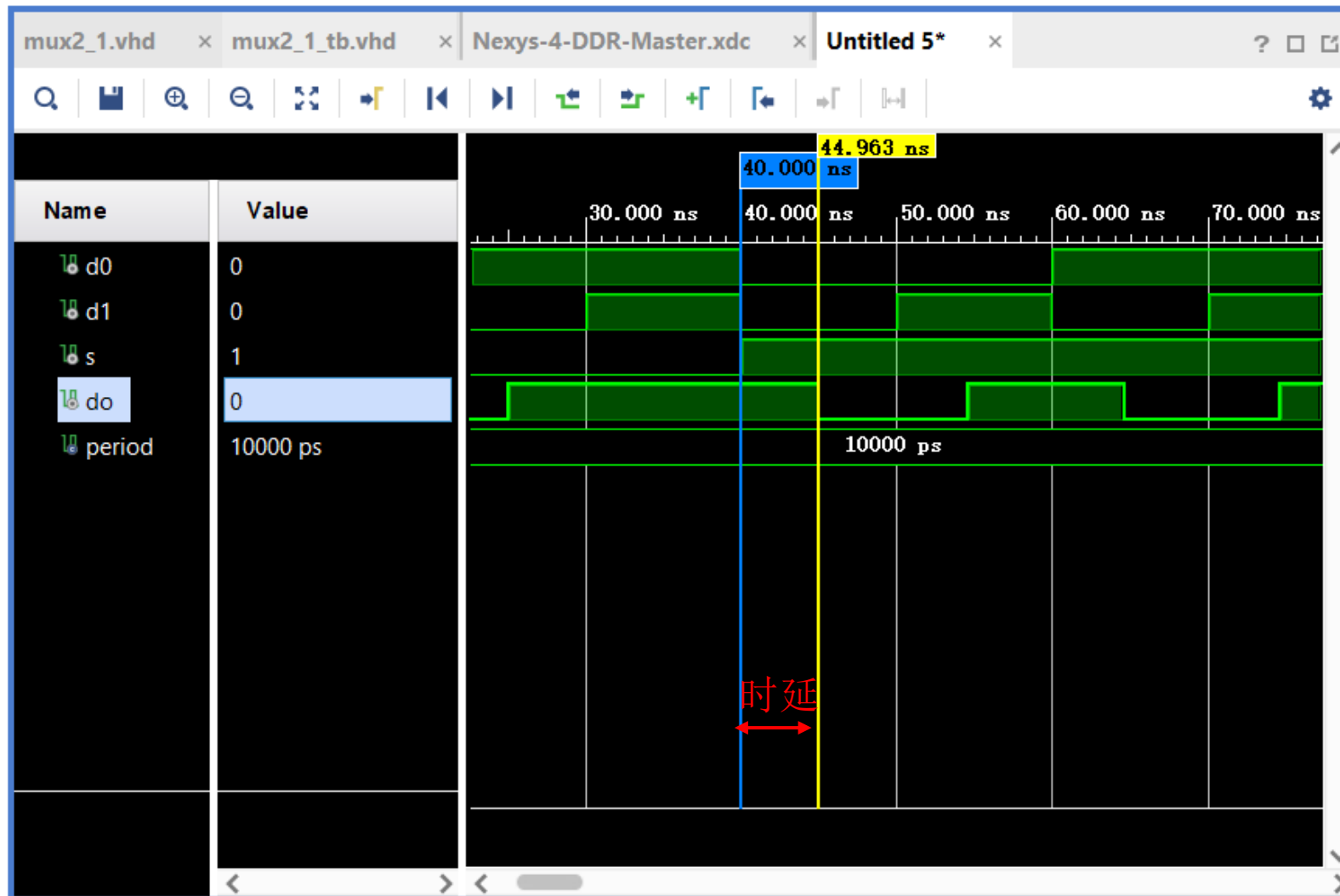
- ▼ IMPLEMENTATION
 - ▶ Run Implementation
 - > Open Implemented Design



Post-Implementation Simulation



Timing Simulation Result

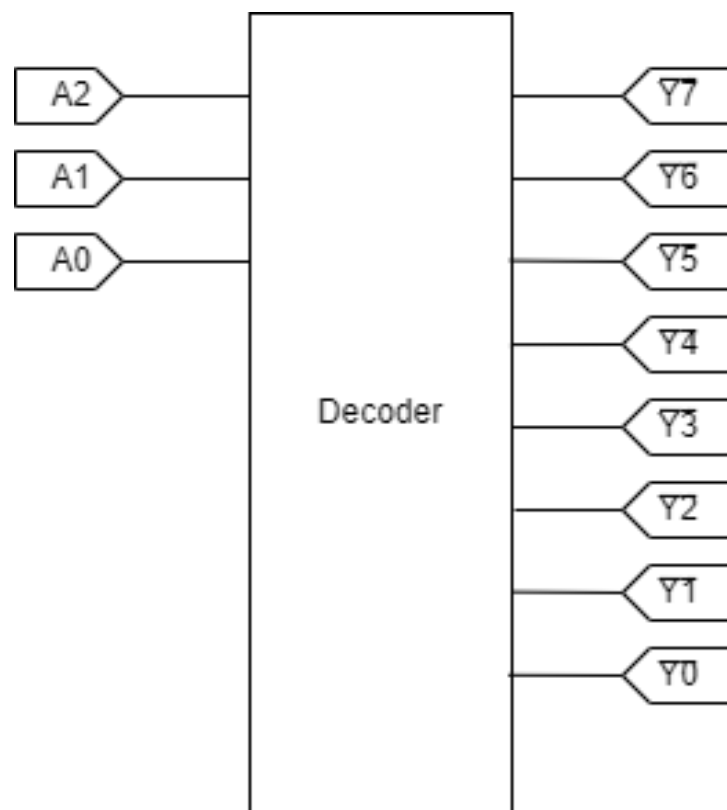


确认正确后即可
进行板级验证

考虑布局布线后时延会有变化

Exercise

请设计一个3-8译码器，编写测试文件对其进行仿真测试。



例如A2~A0为110时，
输出Y7~Y0为1011 1111